

Week 2: Requirements Analysis & System Design

Project: ScientificCalculator

Programming Language: Java

1. Functional Requirements

The Scientific Calculator should provide the following functionalities:

Basic Arithmetic Operations

- Addition (+)
- Subtraction (-)
- Multiplication (×)
- Division (÷)

Scientific Functions

- Sine (sin)
- Cosine (cos)
- Tangent (tan)
- Logarithm (log)
- Natural Logarithm (ln)

Advanced Calculations

- Square (x^2)
- Square Root ($\sqrt{}$)
- Power (x^y)
- Percentage (%)

Display & Utility Features

- Clear, readable display of input and results
- All Clear (AC)
- Clear Entry (CE)
- Backspace
- ANS (Previous Answer)

2. System Workflow Design

Workflow Description

- The user enters a number.
- The user selects an arithmetic or scientific operation.
- The system processes the calculation.
- The result is displayed on the screen.
- The user can continue with another calculation or exit the application.

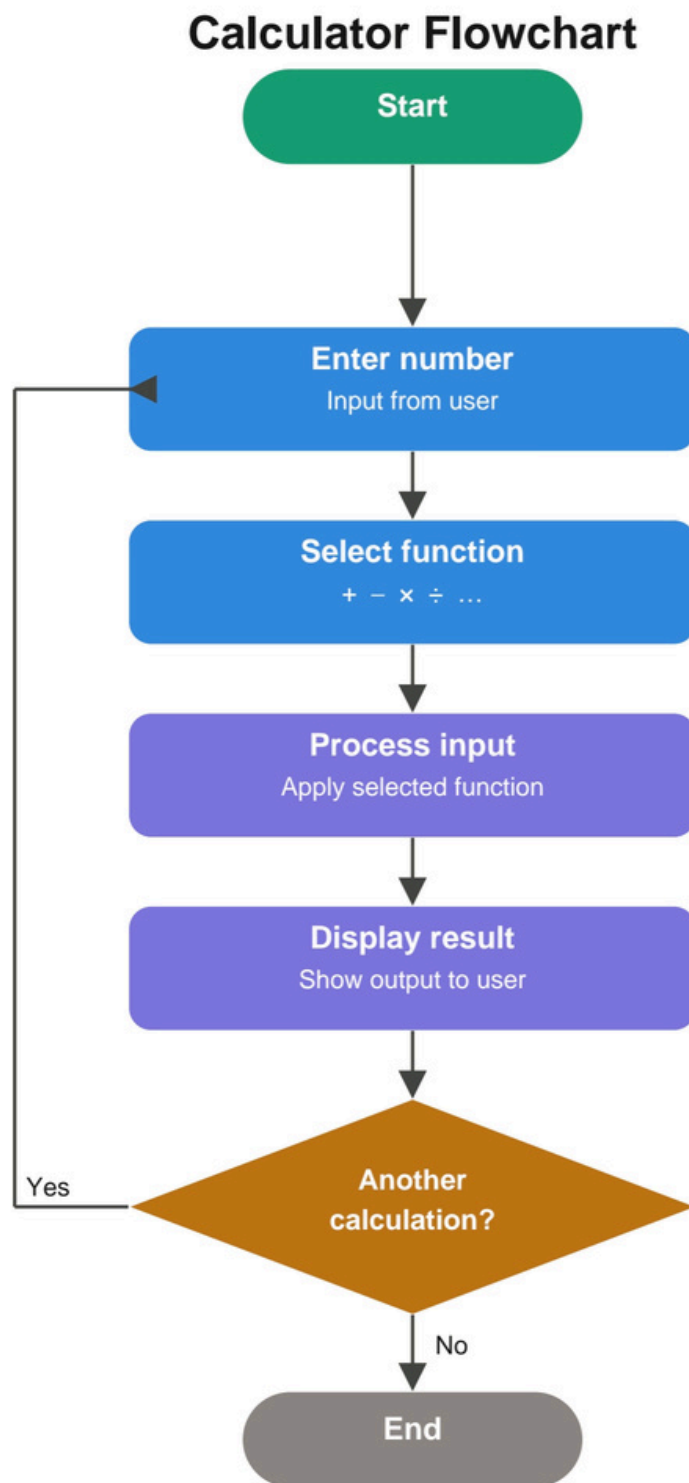


Figure 1 — Scientific Calculator processing workflow

UI Layout Description

The user interface of the Scientific Calculator is designed with a simple and organized layout for easy use.

- Header Section: Displays the calculator title and the ON/OFF button.
- Display Panel: Shows the calculation history, current input, and final result.
- Scientific Buttons: Includes advanced mathematical functions such as sin, cos, tan, log, ln, x^2 , $\sqrt{}$, x^y , $1/x$, %, π , and e.
- Control Buttons: Includes control functions such as AC, CE, Backspace, Parentheses, ANS, and $\pm$.
- Button Grid: Contains numeric buttons (0–9), decimal point (.), and arithmetic operators (+, −, $\times$, $\div$, =).

3. Basic UI Wireframe

The wireframe below outlines the planned layout for the calculator interface, consisting of a header, a result display, a row of scientific function keys, the numeric keypad with basic operators, and a utility row for clearing and advanced operations.

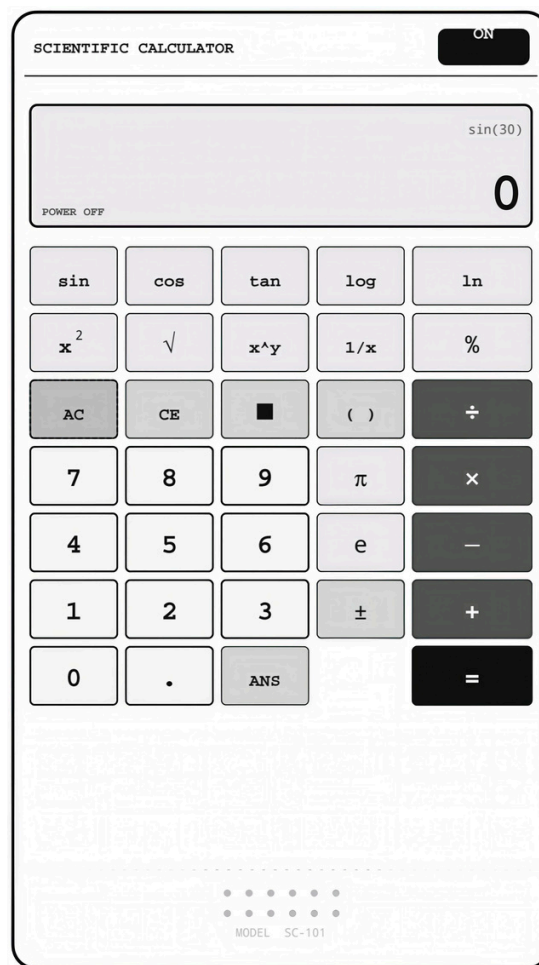


Figure 2 — Basic calculator UI wireframe

Deliverables Completed

- ✓ Functional requirements gathered and documented.
- ✓ System workflow designed.
- ✓ Basic UI wireframe prepared.
- ✓ Requirements analysis completed.